Sanjana Paniyadi

716-533-9781 | Seattle, WA | psanjuupadhyaya@gmail.com | Portfolio | LinkedIn | GitHub

Software Engineer targeting backend and full-stack roles, with 5+ years of experience building scalable web applications, APIs, and data-intensive systems using Java, Python, Node.js, TypeScript, and React. Brings strong backend depth with end-to-end product ownership to deliver reliable, high-performance products, and build systems that scale with business needs.

Education

University at Buffalo Buffalo, NY

Master of Science in Computer Science and Engineering (AI/ML) - 3.74/4 GPA *Aug 2024 – Feb 2026*

Relevant Coursework: Data Structures and Algorithms, Distributed Systems, ML

Experience

Senior Software Engineer | Holiday Channel | Remote Jun 2026 – Present

- Developed a **partner-facing operations platform** using **Next.js and TypeScript**, enabling vendors to manage orders, customers, geographic distribution, and performance metrics across **5+ operational views**.
- Implemented **role-based access control and authorization workflows** to securely isolate data across **multiple vendor accounts** and support permission-based access for **3+ user roles** using TypeScript/Nest.js framework

Early Software Engineer | Kubby(Seed Stage) | Austin, Texas May 2025 – Nov 2025

- Owned end-to-end delivery of a user-facing employee platform, from Figma spec to production, using **React, Javascript and Tailwind CSS**, shipping responsive, WCAG-compliant UI that cut onboarding time by **50%**; sole frontend owner.
- Developed **TypeScript/Node.js** services for an account management platform, modeling **MongoDB** data and implementing business logic for **20+ workflows** across users, orders, transactions, and administrative operations.
- Integrated an **LLM-powered** description assistant into product creation workflows using prompt engineering, surfacing contextual suggestions users could accept, edit, or dismiss, reducing manual data entry by **40%**.

Software Engineer | Zoho Corporation | Chennai, India Jul 2020 – Jul 2024

- Owned the end-to-end implementation of distributed, **cloud-native REST APIs** for internal platform and bulk data within a microservices ecosystem, supporting high-scale data processing of **200,000 records per job** for **public and internal teams**, powered by **Java, Spring Boot, ORM-based persistence** and Kafka-driven workflow
- Optimized bulk data retrieval using **primary-key chunking** and efficient query strategies. Reduced execution time by up to **22%** for large-scale data processing jobs.
- Led the redesign of Zoho CRM's Export UI, expanding export capacity from 3k to 200k records and introducing a configurable builder with filters and custom views from design to production, adopted by **300+ users**.
- Improved **system reliability**, performance and fault-tolerance via callback-based automation workflows for job completion and a **Redis-backed** fallback, reducing job polling by **34%** and supporting monitoring metrics, and SLA-driven recovery.
- Designed and built a **React and Java operations dashboard** used by engineering and operations teams to monitor job queues, investigate failures, and triage production issues, reducing incident response time by **30%**.
- Designed **COQL**, a distributed declarative query execution layer with tree-based query planning for **GraphQL-inspired** nested and relational queries, with OpenAPI/Swagger specifications maintained for all public endpoints.
- Implemented secure cross-module query execution using graph-based path validation, dynamic **SQL joins**, and field-level access controls, supporting nested CRM relationships up to **five levels deep**.

Projects

RAG Pipeline with Hybrid Search: Built a production-oriented RAG system using Python, LangChain, ChromaDB, BM25, OpenAI embeddings, and FastAPI, combining dense and sparse retrieval with rank fusion, reranking, citation verification, and confidence scoring to generate grounded answers over internal documentation.

FIRST LIGHT - NVIDIA Spark Hackathon | Best Use of NemoClaw + OpenShell Winner: Built an offline disaster-triage system on NVIDIA DGX Spark, converting aerial imagery into damage grades, priority rankings, routes, and rescue plans. Ran Nemotron Nano 9B/12B VL, Nemotron Lightning, YOLOv8x, xView2, BGE, vLLM, NemoClaw, and OpenShell locally; achieved 98% person-detection recall and 27s/6-tile batch

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL
Frontend: React, Next.js, HTML, CSS, Tailwind CSS
Backend and APIs: Node.js, Spring Boot, REST APIs, Microservices, OpenAPI/Swagger, GraphQL, Apache Kafka, AWS
Data: PostgreSQL, MySQL, MongoDB, Redis, Relational Data Modeling
AI and Engineering Tools: LangChain, LangGraph, OpenAI APIs, Git, GitLab CI/CD, Maven, Postman, Claude Code